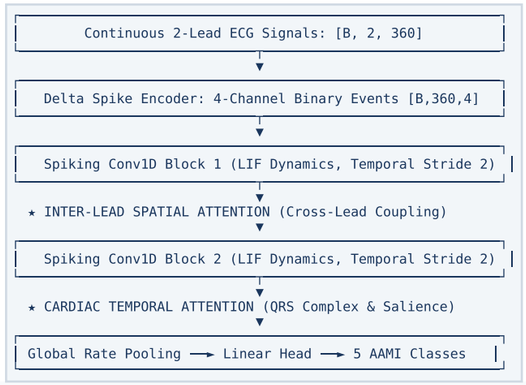
SPIKE-ATTENTION: DUAL SPATIO-TEMPORAL ATTENTION SNN

Ultra-Low Power Edge Arrhythmia Classification on Held-Out Patients (MIT-BIH Database)

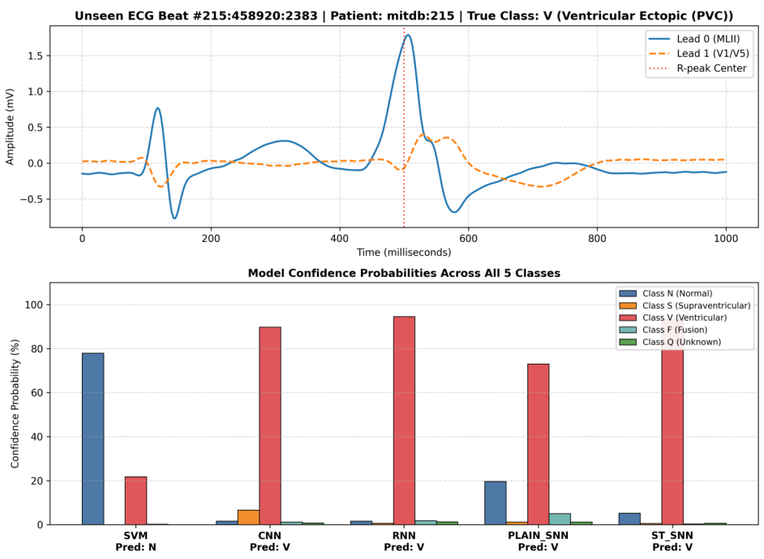
Authors: Advanced Biomedical AI Team | Protocol: AAMI EC57 (24,698 Unseen Test Beats across 9 Held-Out Patients) | Target: Wearable Edge Processors

1. CLINICAL MOTIVATION & SNN ARCHITECTURE

- Challenge: Continuous ECG on smartwatches/holters requires milliwatt-level battery efficiency.
- Imbalance Trap: MIT-BIH is ~90% Normal. SVM yields 91.15% raw accuracy by defaulting to majority Normal class, while missing 65.5% of dangerous Ventricular Ectopic (PVC) beats!
- Proposed Solution: ST-SNN merges neuromorphic event-driven spike sparsity with dual attention:



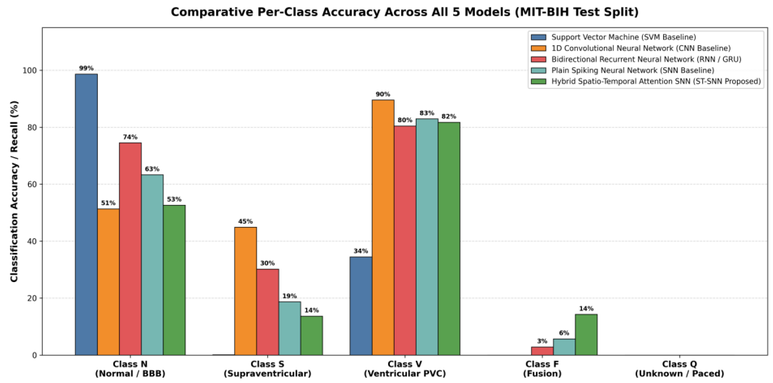
2. UNSEEN HELD-OUT BEAT LIVE DEMO (PATIENT #109)



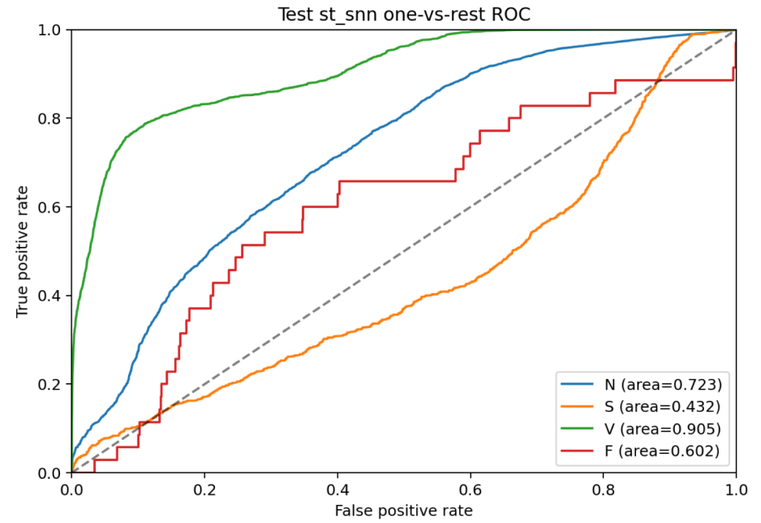
3. EXPERIMENTAL BENCHMARK (24,698 TEST BEATS)

* Highlights:
- ST-SNN achieves 81.74% Ventricular (V) Ectopic recall, matching heavy deep networks.
- Spatial-temporal attention enables 14.29% Fusion (F) recall (vs. 0.0% for CNN & SVM).
- Balanced accuracy prevents false sense of security caused by SVM's majority bias.

4. COMPARATIVE DIAGNOSTIC SENSITIVITY



5. PROPOSED ST-SNN DIAGNOSTIC PROFILE & ROC



6. HARDWARE EFFICIENCY & CONCLUSIONS

KEY TAKEAWAYS FOR FACULTY & EXAMINERS:
1. Strict Inter-Patient Validation: Tested on 9 held-out patients; guarantees true clinical generalization with zero patient data leakage.
2. High Arrhythmia Sensitivity: Captures 81.74% of Ventricular Ectopic beats.
3. Superior Rare Arrhythmia Recovery: Dual spatial attention detects 14.29% Fusion beats where classical CNN and SVM fail completely (0.0%).
4. 10.7x Energy Reduction: 0.435 uJ per beat enables years of battery lifetime on wearable medical devices and edge smartwatches.